FOR SYNTHETIC BIOLOGY RESEARCHERS

WHY LAB SUCCESS DOESN'T GUARANTEE FIELD SUCCESS

Synthetic biology is producing tools directly relevant to problems communities face — water quality, food safety, disease. But the lab-to-field pathway is often built through opportunistic partnerships, and most research happens inside an academic bubble without local context.

Across three case studies, the technologies that reached real communities shared a pattern: local partnership was built alongside the tool, not bolted on afterward.

Grounded in: experience on EWB water infrastructure projects in Central America — sustainable development requires community independence. A solution that requires outside experts to return isn't sustainable.

FIVE DEPLOYMENT READINESS FACTORS

- **Cold chain** — independence from refrigeration
- **Partnership model** — local co-design vs. co-execution
- **Regulatory path** — environmental & political
- **Cost target** — under $5–10 per use for most cases
- **Usability design** — non-expert interpretable output

THREE QUESTIONS TO ASK THROUGHOUT THE PROCESS

1 WHO IS DEFINING THE PROBLEM?

You cannot tell a community what their problem is. The most successful tools here share one trait: the in-country partnership was built alongside the tool.

2 INDEPENDENCE OR DEPENDENCY?

Transferring tools from academic labs is faster. Building local capacity produces communities that can own the science.

3 HOW DOES LOCAL CONTEXT CHANGE MY DESIGN CONSTRAINTS?

Include social scientists in design. Explicitly consider how your design would change outside a well-equipped lab.

From the researchers

"For impactful technology you need to understand the whole ecosystem — technical, regulatory, political."

JULIUS LUCKS · NORTHWESTERN — ROSALIND

"Access to bioreagents is a real limit in the Global South — antibodies and enzymes get stuck in customs, and cost far more once they arrive."

KEITH PARDEE · UNIVERSITY OF TORONTO — PAPER DIAGNOSTICS

A FIELD GUIDE FOR TWO PERSPECTIVES



From lab TO FIELD

HOW SYNTHETIC BIOLOGY CAN SERVE GLOBAL DEVELOPMENT

Ethan Halingstad

SynBAS NRT · Northwestern Center for Synthetic Biology

Engineers Without Borders — Chicagoland Professional Chapter

Two communities, one goal: technology that actually reaches the people who need it.

◄ IF YOU'RE A RESEARCHER, READ THIS SIDE

FLIP OVER IF YOU WORK IN GLOBAL DEVELOPMENT ►

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WHAT CAN SYNTHETIC BIOLOGY ACTUALLY DO FOR YOU?

Synthetic biology is a set of tools for engineering biology to solve practical problems — not just a lab science. All three technologies here were tested in the field, with real communities.

Each case shows what the technology does, what it takes to deploy it, and what a good research partnership looked like from the community side.

CASE 1

ROSALIND: WATER CONTAMINANT TESTING

A paper-based test that detects contaminants like fluoride in drinking water — no lab, no cold storage, and a color readout non-experts can interpret.





98.2%
of 36 Kenyan participants read the test correctly themselves

≈$10
cost per test

WHAT MADE THE PARTNERSHIP WORK

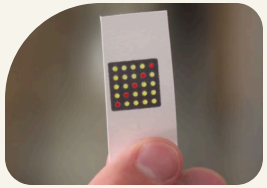
- Kenyan co-investigators embedded from the start
- Usability tested as a core design metric, not an afterthought
- No cold chain and no GMO release — easier regulatory path

CASE 2

PAPER DIAGNOSTICS FOR VIRAL DISEASE

A portable test for viral outbreaks — originally developed for Zika — giving lab-grade accuracy without a PCR machine or a cold supply chain.





98.5%
accuracy vs. CDC gold-standard PCR

$5
per test, used live in the 2015–16 outbreak in Recife, Brazil

WHAT MADE THE PARTNERSHIP WORK

- Fiocruz, the Brazilian public health institute, was a partner from the planning stage — not brought in after the fact
- In-country expertise shaped what the design needed to be
- It later became a startup (En Carta Diagnostics) — a real path from field pilot to lasting local capacity

CASE 3 — READ WITH CARE

GENE DRIVE MOSQUITOES

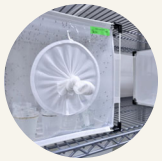
Mosquitoes engineered to resist malaria parasites, designed to spread the trait through wild populations. This is the highest-stakes technology in this guide — an ecosystem-level intervention, not a one-time test.

Two programs show why the partnership model matters as much as the science.



TANZANIA

Engineering done locally at the Ifakara Health Institute, against locally circulating parasite strains, with Tanzanian scientists as co-creators and ongoing community consent — seen as a promising model.



BURKINA FASO

Program suspended in August 2025 amid civil society opposition and sovereignty concerns — the technology-first approach lacked the same local co-ownership.

Field release has not yet occurred for gene drive mosquitoes anywhere. This is presented as an example of the stakes of partnership design, not a ready-to-deploy solution.

HOW TO CONNECT WITH RESEARCHERS

Academic synthetic biology labs are often genuinely looking for real-world problems to work on — but they need practitioners to help define the problem, not just receive the solution.

A good first conversation includes

- A specific, concrete problem — not just "we need better technology"
- What infrastructure is actually available on the ground: power, refrigeration, trained staff, regulatory bodies
- Willingness to co-design from the start, with local staff or scientists as genuine partners — not just field-test hosts
- Realistic timelines — meaningful field-ready technology takes years, not months

This guide was produced as part of an NSF-funded research traineeship project exploring the intersection of synthetic biology and global development practice.